

TRADES-RUNTIME

Architecture & Operating Design Specification

v0.1 - Shadow-First AI Operating System for HVAC, Plumbing & Electrical

Core Principle

Trades-Runtime does not begin by replacing the existing business stack. It first shadows it, records what actually happened, records what Trades would have done, records every human override, then proves or disproves its decisions against operational outcomes.

Target operating scale: 13-15 branches | 750+ employees | Midwest region | multi-trade field service

Primary integration posture: tether to ServiceTitan + ProBooks/other authorized systems first; operate as a read-only or recommendation layer until evidence supports greater autonomy.

This document captures the architecture, ideas, rules, controls and operating logic established for Trades-Runtime.

1. Executive Summary

Trades-Runtime is a company operating intelligence layer designed to coordinate booking, dispatch, technician selection, field execution, warranty logic, pricing, inventory, sales turnovers, service quality, management escalation and continuous operational learning across a large multi-branch trades business.

Its first deployment mode is shadow operation. Existing systems and humans continue running the company. Trades-Runtime observes the same inputs, produces its own decisions, follows the day as reality changes, and compares its counterfactual choices with actual outcomes. The goal is not to claim that the AI would have performed better; the goal is to build call-level and day-level evidence showing when it did, when it did not, and why.

Human Authority Rule

Authorized human decisions always win operationally. Trades-Runtime may recommend against an override and explain why, but it never supersedes authorized human recalibration. The disagreement remains preserved as evidence.

1.1 Primary objectives

- Answer and book customer calls with shared visibility into membership, equipment, warranty and service history.
- Choose the correct technician/Van for the correct call using technical fit, quality, client fit, geography, inventory, runtime, profit, margin and development opportunity.
- Minimize unnecessary travel and fuel through geographic/ZIP consolidation without allowing proximity to become the final deciding factor.
- Continuously learn technician strengths, weaknesses, role suitability and uncertainty without poisoning the model through self-reinforcing assignment bias.
- Use before/during/after photos and video to normalize skill, difficulty, time, workmanship and legitimate added cost.
- Track repairs, turnovers, sales, warranties, callbacks, inventory and downstream revenue as connected events.
- Provide receipts for recommendations, disagreements, overrides, actual outcomes and counterfactual outcomes.
- Allow a field-side AI advisor to be turned on or off per technician/per call.
- Create a canonical live operating state shared across IT, office, dispatch, managers, field technicians and client booking interfaces.

1.2 What Trades-Runtime is not

- Not a simple dispatch optimizer.
- Not a leaderboard that equates revenue with technician quality.
- Not a fixed morning route planner.
- Not an autonomous black box that can silently overrule humans.
- Not an AI that treats a photo inference as proof of fault or misconduct.
- Not a fuel-minimization system that sacrifices technical fit or customer outcome.
- Not a system that assumes a 30-day return visit is automatically a technician callback.

2. System Operating Model

2.1 Canonical live state

Trades-Runtime maintains one canonical live company state. Each role receives a role-specific view of that same state. A completed field event can therefore update technician availability, route position, inventory, booking capacity, management forecasts, warranty context and dispatch recommendations without each department maintaining conflicting versions of reality.

Canonical state inputs include customer, equipment, membership, service history, booking state, technician/Van state, location, inventory, active call state, media evidence, quote state, warranty state, route state, manager queue, pricebook state and integration health.

2.2 Three immutable tracking chains

Chain	Purpose	What it preserves
A. Full-Day Trajectory	Tracks the day as an evolving system	Morning plan, each real-world state change, each rebase, schedule displacement, route drift, end-of-day result
B. Call-to-Call Ledger	Tracks every individual service event	Assignment, travel, arrival, media, diagnosis, parts, quote, repair/turnover, runtime, customer outcome, next-state effect
C. Human Override / Recalibration	Tracks intentional human changes	Original Trades recommendation, role/person who changed it, timestamp, stated reason, replacement decision, downstream effect, final outcome

Never overwrite history

Planned state, actual state, counterfactual state and human override state remain separately queryable. A rebase changes the next decision; it does not erase the prior decision.

2.3 Shadow counterfactual

For each decision point, the system asks: "Given only the information available at that moment, what would Trades-Runtime have done?" This prevents hindsight leakage. Once reality diverges from the recommendation, the shadow model rebases from the new actual state rather than pretending the original imaginary schedule still exists.

The resulting evidence model is:

ACTUAL: customer call -> existing booking/dispatch -> actual Van -> actual diagnosis/quote/repair/turnover -> actual revenue/margin/time/parts.

TRADES: same known inputs -> Trades booking/dispatch -> predicted Van/route -> expected diagnosis/repair/turnover -> expected revenue/margin/time/parts.

OVERRIDE: Trades recommendation -> authorized human change -> reason -> replacement action -> resulting state -> observed outcome.

3. Booking & Schedule Precedence

3.1 Booking precedence rule

Already-booked calls establish precedence. The default policy is to preserve customer commitments rather than optimize the schedule by repeatedly moving customers for marginal route gains.

Schedule rule

Preserve existing customer commitments unless the expected operational or customer consequence of preserving them exceeds the cost of displacement.

3.2 Demand-based displacement

Demand service can justify rescheduling when the service need materially outweighs the protected appointment. Displacement is never silent. Trades-Runtime evaluates alternatives, documents why a move is recommended, estimates customer impact, and requires the appropriate operational workflow for customer contact/approval.

- Urgency and safety impact.
- Membership/service-level commitments.
- Existing appointment precedence and promised window.
- Available technician competence and inventory.
- Travel and current route position.
- Expected call runtime and downstream schedule damage.
- Likelihood of successful rebooking or cancellation.
- Revenue/contribution consequences, but never revenue alone.
- Customer history, prior disruptions and management sensitivity.

4. Dispatch Intelligence

4.1 Call-Fit Score

Every eligible Van receives a dynamic multi-objective Call-Fit Score. It is not a single sales metric and it is not distance-first.

Call-Fit = quality + service competence + technical fit + client-tech fit + geolocation + truck inventory + estimated runtime + expected fixed-dollar contribution + expected margin + schedule impact + exploration opportunity.

4.2 Geographic and ZIP intelligence

The system learns geographic clusters, ZIP patterns, branch boundaries, traffic/travel behavior and common service corridors. Vans can be preferentially positioned or routed to consolidate travel and reduce fuel cost, but geographic efficiency is never the final deciding factor. A farther technician can win when expected technical resolution, inventory coverage, quality or contribution materially outweighs the route savings.

4.3 Dynamic route drift

An 8-12 hour field day will not remain on the route imagined at the start of the day. Trades-Runtime treats drift as normal operating evidence. After every meaningful event, the remaining schedule is recalculated from the actual present state.

Morning plan -> event -> actual position/state -> rebase -> next event -> rebase -> close.

Key day-level measures are morning forecast quality, adaptation quality and final-day operating outcome.

4.4 Runtime modeling

Trades-Runtime learns standard call duration by issue, trade, equipment type, access/difficulty profile and technician. Generic time blocks are progressively replaced by distributions learned from actual work. Runtime influences not only the current assignment, but the expected feasibility of the next calls in the day.

4.5 Exploration / curveball mechanism

Pure exploitation creates a poisoned model: technicians who historically receive a job type gain more evidence and opportunity while others appear weak because they are never tested. Trades-Runtime therefore maintains an explicit exploration mechanism.

- Exploration is controlled and competence-gated, not random assignment.
- The system tracks observed capability separately from uncertainty/lack of opportunity.
- Appropriate technicians can receive selected new opportunities when adjacent skills suggest they can safely perform them.
- Exploration results update the technician skill model and reduce uncertainty.
- Long-term workforce development can justify a small short-term optimization tradeoff when risk remains acceptable.

5. Technician / Van Intelligence

5.1 Identity and privacy model

Operational analytics should primarily reference Van identifiers rather than technician names. Identity can remain available to authorized systems/roles, but optimization and reporting can be pseudonymized where practical.

5.2 Editable internal role tags

Tag	Operational meaning
Service 1	Entry/basic service capability tier
Service 2	Intermediate service capability tier
Service 3	Advanced service capability tier
Warranty	Technician optimized/approved for warranty work
Selling Tech	Equipment-sales capable field technician
Repair Tech	Technician oriented toward complex repair execution
Comfort Advisor	Dedicated sales/comfort consultant
Manager	Authorized management/escalation role

Tags are fully modifiable by authorized humans. Trades-Runtime can recommend a change, recommend against a change, and explain its evidence, but cannot supersede authorized human recalibration. Every modification lands in the override/recalibration chain.

5.3 Multi-horizon technician position

Technician position must be shown across multiple time horizons to prevent a short spike or slump from becoming the employee narrative.

Window	Use
1 Week	Immediate operating trend and recent behavior
30 Days	Current performance profile
60 Days	Near-term stability/change
90 Days	Quarter-scale capability and consistency
6 Months	Structural trend and longer-term position

5.4 Output vs performance

Output and performance are separate. Output reports what happened; performance estimates how well it happened after context normalization.

Output	Performance
Calls completed	Quality/workmanship
Repair dollars	Callback-adjusted competence
Turnovers created	Diagnostic accuracy
Turnovers sold	First-trip completion
IAQ/enhancement sales	Runtime efficiency after difficulty adjustment
Memberships	Client-tech fit
Equipment sales attribution	Margin quality
Gross contribution	Manager intervention / escalation pattern

6. Visual & Video QA Engine

6.1 Media as field ground truth

Before, during and after photos and video are primary calibration evidence. Media is combined with notes, equipment information, parts, elapsed time, diagnostic readings, quotes, callbacks and final system performance. Media inference alone is never treated as definitive proof of fault, misconduct or workmanship failure.

6.2 What the engine estimates

- Initial equipment condition and equipment age indicators.
- Access difficulty, corrosion, crowding, contamination, wiring complexity, installation condition and other effort drivers.
- Likely repair difficulty and expected labor time.
- Workmanship, completeness and visible quality after repair.
- Expected parts/BOM and whether the installed/used parts are consistent with the scope.
- Legitimate added-cost drivers tied to unusual access, condition, procurement or complexity.
- Effort required versus effort demonstrated.
- Potential concerns visible in media that are absent from technician notes, flagged as review items rather than accusations.
- Callback risk and documentation sufficiency.

6.3 Skill normalization

Visual/video evidence helps distinguish a slow technician from a difficult repair and distinguishes high revenue from high technical competence. A technician who completes a severe-access repair correctly in 150 minutes can outperform a technician who completes an easy repair in 60 minutes if the expected normalized time and quality support that conclusion.

7. Customer, Equipment, Warranty & Callback Intelligence

7.1 Required context

- Equipment age and installation date when known.
- Model/serial and registration status.
- Parts warranty status.

- Labor warranty status.
- Membership status and entitlements.
- Last visit and prior service history.
- Prior diagnoses, quotes and declined work.
- Prior repairs and parts used.
- Photos/video and documented equipment condition.

7.2 Warranty lookup via MCP/connectors

Authorized connectors/MCP tools can query manufacturer, distributor, internal or other warranty sources for parts and labor coverage. Coverage should be exposed during booking, dispatch, field diagnosis, quoting and repair reconciliation. Warranty states should include CONFIRMED, NOT COVERED, POSSIBLE, UNKNOWN and LOOKUP FAILED so an unavailable lookup never silently becomes "not covered."

7.3 30-day callback classification

A return visit inside 30 days is a trigger for review, not an automatic technician penalty. The classifier asks what failed, what was previously observed, what was quoted, what was accepted or declined, and what work the prior technician actually performed.

Classification	Interpretation
True callback / workmanship or diagnostic miss	Evidence reasonably ties the repeat problem to prior work or a material miss
Previously quoted / declined	Failed item was identified and customer declined the recommended repair
New unrelated failure	New issue occurred inside the window but is not materially connected
Warranty event	Covered part/labor issue without technician fault
Indeterminate	Evidence insufficient; route for review rather than auto-penalty

8. Manager Intervention Queue

Trades-Runtime continuously estimates where management involvement is likely to improve outcome before a field situation becomes an escalation.

- Repeat visits or unresolved history.
- Angry/escalating customer language.
- Disputed warranty or membership status.
- High-dollar or unusually complex repair decisions.
- Prior declined work now implicated in a failure.
- Technician uncertainty or repeated Field Advisor use.
- Media/notes conflict or documentation gaps.
- Unusually high quote variance.
- Potential callback ambiguity.
- Schedule displacement with material customer impact.

The manager queue is a recommendation and prioritization surface. It does not remove field/office judgment.

9. Field Advisor

9.1 Per-tech/per-call on/off control

The field AI advisor can be enabled or disabled by authorized policy or by the technician/call workflow. When disabled, Trades-Runtime can continue passive observation where permitted. When enabled, it becomes an active diagnostic and repair copilot.

9.2 Field context available to the advisor

- Customer and equipment history.
- Prior repair/quote/decline history.
- Parts/labor warranty and membership status.
- Known model/serial information.
- Current truck inventory.
- Pricebook matches.
- Photos/video/readings/notes supplied during the call.
- Likely failure hierarchy and next diagnostic checks.
- Predicted parts, runtime, difficulty and procurement needs.
- Suggested repair/quote options and customer explanation.

9.3 Learning from advice usage

The system can measure whether AI guidance was requested, followed or rejected and what outcome followed. This is not a simplistic surveillance score. The purpose is to learn where assistance improves outcomes, where technicians outperform the model, and where the model needs recalibration.

10. Pricebook & Profitability Engine

10.1 ServiceTitan pricebook adaptation

Trades-Runtime ingests the authorized pricebook and compares current pricing with observed repair volume, parts cost, labor burden, difficulty, access, procurement friction, callback risk, demand, market constraints and target gross contribution.

Price recommendations should consider both fixed-dollar contribution and percentage margin. High-volume work can have different optimal economics than rare, high-complexity work.

10.2 Quote-from-field function

One action should be able to combine technician notes, photos/video, equipment context, predicted parts, difficulty, access, procurement, inventory, warranty and the pricebook into a proposed quote package with confidence and assumptions.

- Probable repair/scope.
- Expected parts/BOM.
- Part availability and procurement path.
- Expected labor/runtime.
- Difficulty/access modifiers.
- Warranty effect.
- Pricebook item mapping.

- Expected revenue, fixed-dollar gross contribution and margin.
- Confidence plus evidence receipts.

11. Truck Inventory Intelligence

Truck inventory is dynamic and technician-specific. Trades-Runtime compares what it predicted would be required with what ServiceTitan/other systems show was actually consumed, then adjusts recommended stock by technician skill profile, territory, seasonal call mix and likely future work.

- Inventory fit affects dispatch because it changes first-trip completion probability.
- Actual consumption reconciles the prediction model.
- Frequent emergency procurement becomes evidence of a stocking gap.
- Stock recommendations should consider carrying cost and vehicle capacity, not simply maximize inventory.
- Parts prediction performance becomes a measurable model-quality metric.

12. Revenue, Repair & Sales Attribution

Trades-Runtime keeps the originating service call connected to all downstream outcomes rather than separating service and sales into unrelated systems.

- Functional repair revenue.
- IAQ revenue.
- Enhancements/accessories.
- Warranty repairs.
- Memberships.
- Equipment turnovers generated.
- Turnovers sold by Comfort Advisors/sales staff.
- Gross contribution and margin.
- Callback or rework cost.
- Attributable downstream revenue from the original service interaction.

This supports fairer analysis of who identifies opportunities, who closes them, and what combination of technician/customer/call/manager/route conditions produces the strongest company outcome.

13. Human Override & Recalibration Governance

13.1 Universal override

Any automated or recommended decision can be manually overridden by an authorized human within that role's permissions. Areas include booking, dispatch, route, technician assignment, schedule displacement, warranty classification, role tags, quote/price suggestions, parts, inventory and escalation.

13.2 Override receipt

Each override records:

- Original Trades recommendation and confidence.
- Who/which authorized role overrode it.
- Timestamp.
- Stated reason.

- Replacement decision.
- Immediate state change.
- Downstream schedule/route effects.
- Observed final outcome.
- Later assessment of whether the override improved, harmed or had neutral effect.

Learning rule

A human who repeatedly beats the model is not "noncompliant." The system should search for the unmodeled signal that person is using and incorporate it when valid.

14. Company Heartbeat & Role Interfaces

Trades-Runtime acts as the shared nervous system between client booking, office staff, dispatch, field technicians, managers and IT. Every role sees only the information/actions appropriate to it, but all views are generated from the same canonical state.

Role	Primary view/actions
Client Booking	Customer identity, membership, equipment/history, available windows, urgency, appointment precedence
Office	Booking state, customer issues, warranty/membership context, schedule changes, contact actions
Dispatch	Live Van state, route, Call-Fit, runtime, inventory fit, alternatives, override controls
Field Tech	Call context, Field Advisor, history, warranty, inventory, quote/parts support, media capture
Manager	Escalation queue, override/recalibration, technician trends, quality, revenue and risk
IT / System Admin	Integration health, MCP/API status, permissions, event integrity, model health, audit logs

Heartbeat updates should be event-driven where possible, with periodic health checks so a stale integration or missing field event is visible rather than silently corrupting the operating picture.

15. Receipt Architecture

Every important recommendation should be explainable after the fact. A receipt is the minimum evidence packet required to reconstruct why the system believed an action was appropriate at the time.

- Input snapshot and timestamp.
- Known customer/equipment/warranty/membership state.
- Candidate Vans/alternatives considered.
- Scoring factors and confidence.
- Chosen recommendation.
- Human/system actual decision.
- Override information if applicable.
- Predicted outcome and expected uncertainty.
- Actual outcome.
- Counterfactual assessment.
- Which model parameters or technician profile elements changed afterward.

16. Shadow-Mode Proof Framework

16.1 Deployment ladder

1. Observe - ingest and normalize data without influencing decisions.
2. Shadow - produce sealed counterfactual recommendations while humans/existing software operate normally.
3. Recommend - surface decisions to staff with no autonomous execution.
4. Human Approves - one-click acceptance/override workflow.
5. Limited Autonomy - narrowly scoped actions with hard constraints and rollback.
6. Autonomous Operations - only after sustained evidence, governance and operational approval.

16.2 Proof metrics

Trades-Runtime should not claim success using revenue alone. Shadow proof should include:

- Gross contribution delta and margin delta.
- First-trip completion.
- Callback/rework rate.
- Diagnostic accuracy.
- Turnovers created and sold.
- Travel/fuel/time efficiency.
- Customer wait and displacement impact.
- Runtime forecast error.
- Inventory prediction accuracy.
- Warranty/callback classification accuracy.
- Quality/workmanship indicators.
- Manager intervention prediction quality.
- Human override win/loss/neutral rate.
- Exploration results and newly discovered technician capability.

17. Core Data Objects

Object	Minimum responsibility
Customer	Identity, address/service location, membership, preferences, communication history
Equipment	Trade, type, model/serial, age, install date, warranty, service history
Call	Issue, urgency, booking, required skills, precedence, lifecycle state
Van/Technician	Role tags, skills, quality, location, inventory, schedule, uncertainty/exploration state
Media Evidence	Before/during/after photos/video, timestamps, call linkage, findings/confidence
Quote/Repair	Diagnosis, scope, parts, pricebook mapping, labor, difficulty, margin, warranty
Inventory Event	Predicted need, stock, consumption, replenishment/procurement
Turnover/Sale	Originating call, tech, sales role, close, amount, contribution
Trajectory Event	Planned state, actual event, rebase, new schedule/route state
Override Event	Recommendation, authorized human change, reason, replacement, outcome
Receipt	Reconstructable evidence for any material system recommendation

18. Decision Rules to Freeze in v0.1

1. Human-authorized recalibration always controls the live operation.
2. No recommendation may erase the recommendation or state that preceded it.
3. Route drift is expected and causes rebasing, not automatic failure scoring.
4. Existing bookings have precedence; displacement must be justified and auditable.
5. Geography/fuel matters but is never the final factor.
6. Technician revenue is not equivalent to technician skill.
7. Skill must be normalized for call difficulty, access, runtime, opportunity mix and customer context.
8. Inside-30-day return visits are reviewed, not automatically charged as callbacks.
9. Previously quoted and declined work must be separated from technician fault.
10. Media can support a concern but cannot alone prove misconduct or diagnostic failure.
11. Unknown warranty status is never silently converted to not-covered.
12. Exploration opportunities are necessary to prevent historical dispatch patterns from poisoning capability estimates.
13. Output and performance remain separate metrics.
14. Trades-Runtime must be able to explain material decisions with receipts.
15. Actual real-world state always becomes the starting point for the next decision.

19. Recommended Logical Architecture

A practical high-level architecture is:

ServiceTitan / ProBooks / warranty sources / telephony / maps-location / pricebook / inventory / media -> Integration & MCP Layer -> Canonical Event Bus -> Operational State Store -> Decision Engines -> Shadow/Counterfactual Engine -> Receipts & Audit -> Role Interfaces -> Controlled Action Layer.

19.1 Decision engines

- Booking & Call Intake Engine.
- Dispatch / Call-Fit Engine.
- Full-Day Trajectory Engine.
- Runtime Prediction Engine.
- Technician Skill & Exploration Engine.
- Visual/Video QA Engine.
- Warranty & Callback Classifier.
- Manager Intervention Predictor.
- Field Advisor.
- Pricebook / Quote / Margin Engine.
- Inventory Prediction Engine.
- Revenue & Turnover Attribution Engine.
- Counterfactual Outcome Engine.
- Receipt / Audit / Override Engine.

19.2 Data/storage separation

- Immutable event ledger for audit and historical reconstruction.
- Operational state store for the current live company state.

- Analytical warehouse/lakehouse for long-horizon training and reporting.
- Media/object store for photos/video with strict call/equipment linkage.
- Feature/skill store for technician and job-type models.
- Receipt store for decision evidence and override history.

20. Safety, Fairness & Governance Requirements

- Role-based permissions for sensitive employee/customer data and overrides.
- Pseudonymized Van-first analytics where practical.
- No automatic disciplinary action from a model score, media inference or one time window.
- Performance views must show context and uncertainty, not only rank.
- Photo/video analysis should be used for work quality and job context within authorized policy, with retention/access controls.
- Customer/employee recording and media workflows must comply with applicable consent, privacy and employment requirements.
- Every autonomous capability must have a manual fallback and clear integration-health state.
- External API/MCP integrations must be validated against actual permissions, rate limits, data rights and available endpoints before implementation.

21. v0.1 Build Boundary

The first build should favor proof over autonomy. The most valuable v0.1 is a reliable shadow operating system that can reconstruct calls, technicians/Vans, routes, booking decisions, warranty/membership context, outcomes and overrides with defensible receipts.

21.1 Minimum viable shadow system

1. Ingest ServiceTitan + ProBooks/authorized operational data.
2. Normalize customers, equipment, calls, Vans/technicians, bookings, jobs, quotes, parts, turnovers and sales.
3. Create canonical event timestamps and call IDs across integrations.
4. Map technician identities to Van identifiers for primary analytics.
5. Implement Full-Day Trajectory, Call-to-Call Ledger and Override/Recalibration Chain.
6. Implement baseline Call-Fit scoring and route rebasing.
7. Implement technician role tags and multi-window dashboards (1 week / 30 / 60 / 90 days / 6 months).
8. Implement 30-day callback/warranty decision support.
9. Implement warranty lookup adapters where authorized.
10. Implement media ingestion and initial before/after visual QA scoring.
11. Implement receipts that make every shadow recommendation reconstructable.
12. Compare actual versus shadow outcomes without allowing future information into the original recommendation.

21.2 Subsequent capabilities

Once the shadow data is trustworthy, add live Field Advisor, video reasoning, manager queue, quote generation, pricebook optimization, truck inventory prediction, customer-facing AI call handling and progressively controlled execution.

22. Success Condition

Trades-Runtime succeeds when management can open any day, any call or any meaningful system disagreement and see: what the company knew at that moment, what existing operations did, what Trades-Runtime recommended, whether a human changed it, why, how the remaining day adapted, what physically occurred in the field, what the customer received, and what measurable result followed.

North Star

The system earns authority by being auditable, adaptable and repeatedly correct under real operating conditions - not by being more confident than the humans running the business.

End of architecture specification - Trades-Runtime v0.1